6CCSACCM
Cooperation, Coordination and Multi-Agent Systems

Week 10

Voting (part I)

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Computational social choice and voting theory

- Field at the intersection of algorithms, game theory, and social choice
- Studies how groups of agents make collective decisions in a principled way
- Analyses how voting rules aggregate individual preferences into a single outcome
- Explores paradoxes and impossibility results (e.g., Arrow, Gibbard–Satterthwaite)
- Investigates how computational complexity affects what is feasible in practice
- Shows how complexity can sometimes protect systems from strategic manipulation
- Uses algorithmic techniques to design better rules (elections, committees, etc.)
- Applies beyond political voting: recommender systems, resource allocation, multi-agent AI decision-making

Applications: Recommender systems

- Use voting rules to aggregate user ratings or implicit feedback.
- Construct recommendation lists that represent diverse user groups, not just the average or most vocal users.
- Support group recommendations (e.g., for families, teams) via preference aggregation.

Applications: Resource allocation

- Apply voting/committee rules to choose subsets of projects in participatory budgeting or public-good selection.
- Use preference aggregation to allocate limited resources (computational, physical, or budgetary) fairly across stakeholders.
- Capture trade-offs between efficiency, fairness, and representation in allocation outcomes.

Applications: Multi-agent AI decision-making

- Let multiple agents or models "vote" over actions, plans, or hypotheses to reach a joint decision.
- Use social choice rules to coordinate behaviour in swarms, robot teams, or distributed systems.
- Provide interpretable aggregation of heterogeneous objectives or preferences within a multi-agent architecture.

Opening example

Five voters express their preferences over three alternatives. We need to find a good ranking of the alternatives to reflect this information:

Voter 1	$a \succ b \succ c$
Voter 2	$b \succ c \succ a$
Voter 3	$c \succ a \succ b$
Voter 4	$c \succ a \succ b$
Voter 5	$b \succ c \succ a$
	?

Setting

- Set of alternatives (candidates) $C = \{c_1, \ldots, c_m\}$
  - Represents the possible outcomes or options among which society must choose
- Set of agents (voters) $N = \{1, \ldots, n\}$
  - Represents the agents (people, organisations, or AI systems) who hold preferences over the alternatives
- Voter preferences
  - Each voter has a preference relation over the alternatives (e.g., a ranking, approval set, or scores)
  - Collectively, these form a preference profile that serves as the input to the voting rule
- Aggregation into a single outcome
  - A voting rule takes the preference profile and returns a collective decision
  - The outcome can be a single winner, a set of winners, or an ordering of all alternatives

Preference formats: Rankings

- Each voter orders all (or many) alternatives from most to least preferred
- Used by many traditional voting rules (e.g., Borda, STV, Condorcet methods)
- Capture rich ordinal information but can be costly to elicit when there are many options

Preference formats: Approval sets

- Each voter marks a subset of alternatives as "approved" (acceptable), leaving others unapproved
- Underlie approval voting and many multiwinner rules based on approval
- Simpler to elicit than full rankings and natural when voters face large candidate sets

Preference formats: Scores/stars

- Each voter assigns a numerical rating (e.g., 1–5 stars) or score to each alternative
- Common in recommender and review systems, and in score-based voting rules
- Provide cardinal information about the intensity of preference, enabling aggregation via sums, averages, or other scoring procedures

Outcome formats

- Single-winner rules (social-choice functions)
  - Output exactly one winning alternative
  - Examples include plurality, Borda, and instant-runoff voting
  - Often used in presidential or mayoral elections
- Multiwinner rules
  - Output a committee or a set of several winners of fixed size
  - Used in settings like parliaments, councils, or top-k recommendations
  - Capture goals such as proportional representation or diversity
- Social-welfare functions
  - Output a complete ranking (or ordering) of all alternatives
  - Represent "society's" preference order rather than just a winner/committee
  - Useful when downstream rules or decisions will operate on the social ranking

Single-winner elections

- Fix a finite set $C = \{c_1, \ldots, c_m\}$ of candidates, with $|C| = m \geq 2$.
- Let $\mathcal{L}(C)$ denote the set of all strict linear orders R on C. We use elements of $\mathcal{L}(C)$ to model (true) preferences and (declared) ballots.
- Each member i of a finite set $N = \{1, \ldots, n\}$ of voters supplies us with a ballot R_i, giving rise to a profile $R = (R_1, \ldots, R_n) \in \mathcal{L}(C)^n$.
- A voting rule (or social-choice function) for N and C selects (ideally) one or (in case of a tie) more winners for every such profile:

$$F : \mathcal{L}(C)^n \to 2^C \setminus \{\emptyset\}$$

  If $|F(R)| = 1$ for all profiles R, then F is called *resolute*.
- Most natural voting rules are irresolute and have to be paired with a *tie-breaking* rule to always select a unique election winner (e.g.,random or lexicographic tie-breaking).

Voting rules

Plurality

Definition

Plurality: elect the alternative ranked first most often (i.e., each voter assigns 1 point to an alternative of her choice, and the alternative receiving the most points wins).

- Plurality is *anonymous*: only the multiset of votes matters, not the voters' identities
- Plurality is *neutral*: candidates' identities do not matter (unless we use a non-neutral tie-breaker)
- For 2 candidates, plurality selects the majority winner
- However, for 3 candidates it may behave in an undesirable way (e.g., the majority of voters may prefer some other alternative to the current winner)

Example

Consider plurality among the following ranking preference profile:

1	2	3	4	5
a	a	b	c	d
b	b	c	b	b
c	c	d	d	c
d	d	a	a	a

Borda count

Definition

Borda count: each voter gives m-1 points to the alternative she ranks first, m-2 to the alternative she ranks second, etc.; and the alternative with the most points wins.

- Borda rule improves on plurality by using full rankings rather than just first-place votes, rewarding candidates who are broadly acceptable across voters.

Example

Consider running Borda count on the following preference profile, resulting in the points per alternative shown to the right:

1	2	3	4	5
a	a	b	c	d
b	b	c	b	b
c	c	d	d	c
d	d	a	a	a

$\rightarrow$

alternatives	points
a	6
b	11
c	8
d	5

Positional scoring rules

Definition

A **positional scoring rule (PSR)** is defined by a *scoring vector* $s = (s_1, \ldots, s_m) \in \mathbb{R}^m$ with $s_1 \geq s2 \geq \ldots \geq s_m$ and $s_1 > s_m$. Each voter submits a ranking of the m alternatives. Each alternative receives s_i points for every voter putting it at the ith position. The alternative(s) with the highest score (sum of points) win(s).

- Borda rule = PSR with scoring vector $(m-1, m-2, \ldots, 0)$
- Plurality rule = PSR with scoring vector $(1, 0, \ldots, 0)$
- Veto rule = PSR with scoring vector $(1, \ldots, 1, 0)$
- For any $k < m$, k-approval = PSR with $(1, \ldots, 1, 0, \ldots, 0)$

Elimination-based voting methods

Definition

Plurality with runoff: run a plurality election and retain the two front-runners; then run a majority contest between them.

Definition

Instant-Runoff Voting (IRV): votes are tabulated in rounds, where in each round, the alternative with the lowest plurality score is eliminated; the last alternative left standing is the winner.

Examples of countries and regions using IRV include:

- Ireland: Used for all public elections.
- Canada: Used in Ontario for municipal elections.
- Australia: Used for parliamentary elections.
- USA: Used for statewide elections in Maine (ME) and Alaska (AK), and in cities like NYC and Cambridge.

Example

Consider running IRV on the following preference profile:

1	2	3	4	5
a	a	b	c	d
b	b	c	b	b
c	c	d	d	c
d	d	a	a	a

$\rightarrow$

1	2	3	4	5
a	a	c	c	d
c	c	d	d	c
d	d	a	a	a

$\rightarrow$

1	2	3	4	5
a	a	c	c	c
c	c	a	a	a

Condorcet paradox

The preferences of the majority may be cyclical, meaning no clear winner.

1	2	3
a	c	b
b	a	c
c	b	a

Condorcet principle

Definition

A **Condorcet winner** is an alternative that defeats every other alternative in a head-to-head comparison.

Definition

A rule is **Condorcet-consistent** if it always selects a Condorcet winner whenever it is presented with a profile that contains one.

- Positional scoring rules and elimination-based methods are *not* Condorcet-consistent.

Example

Plurality and IRV are not Condorcet-consistent:

1	2	3	4	5
a	a	b	c	d
b	b	c	b	b
c	c	d	d	c
d	d	a	a	a

Example

Borda count is also not Condorcet-consistent:

1	2	3
a	a	b
b	b	c
c	c	d
d	d	a

$\rightarrow$

alternatives	points
a	6
b	7
c	4
d	1

Condorcet extensions: Copeland (Llull)

Definition

Copeland (Llull's) rule: each alternative receives one point for each head-to-head comparison it wins, and half a point for tied comparisons; the candidate with the highest score wins.

- In an m-candidate election, if a Condorcet winner exists, it gets $m-1$ point, all other candidates get at most $m-2$ points.

Example

Consider Llull's Rule on the following preference profile:

1	2	3	4	5
a	a	b	c	d
b	b	c	b	b
c	c	d	d	c
d	d	a	a	a

$\rightarrow$

alternatives	points
a	0
b	3
c	2
d	1

Condorcet extensions: Maximin

Definition

Maximin rule: the score of each candidate is the number of votes it gets in its worst pairwise election; the candidate with the highest score wins.

- In an n-voter election, if a Condorcet winner exists, its maximin score is greater than $n/2$, while everyone else's maximin score is less than $n/2$.

Condorcet extensions: Dodgson

Definition

Dodgson's rule: the Dodgson score of an alternative x is the minimum number of swaps between adjacent alternatives needed to make x a Condorcet winner; we then select an alternative with the minimum score.

- Note that Dodgson's rule is trivially Condorcet-consistent because a Condorcet winner would have a score of 0 and all other alternatives would have a positive score, as they are not Condorcet winners.
- Additionally, note that Dodgson's rule is NP-hard to compute.

Example

We will calculate the Dodgson score of b in the following preference profile:

1	2	3	4	5
a	a	d	d	d
b	b	c	c	c
c	c	a	b	b
d	d	b	a	a

Here, b is losing in head-to-head comparisons with all other preferences.

Example

We start by flipping b and c in voter 4's preferences to get:

1	2	3	4	5
a	a	d	d	d
b	b	c	b	c
c	c	a	c	b
d	d	b	a	a

Now, b is beating c in a pairwise comparison, but b is still losing to a and d.

Example

We will now swap b and d in voter 4's preferences to get:

1	2	3	4	5
a	a	d	b	d
b	b	c	d	c
c	c	a	c	b
d	d	b	a	a

Now, b is beating c and d in pairwise comparisons but is still losing to a.

Example

So, we swap a and b in voter 1's rankings to get:

1	2	3	4	5
b	a	d	b	d
a	b	c	d	c
c	c	a	c	b
d	d	b	a	a

Now, b is beating all other alternatives according to the preferences of the majority, and so is a Condorcet winner in this profile. As this took 3 swaps (which is the minimum number of swaps) the Dodgson score of b is 3.

THANK YOU!